



How Language Design Affects Human's Mind

-Ruby vs. Virtual Language in Nineteen Eighty-Four-

Composed By: Hayato Ishida

For Japanese version, click [here](#)

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About Me

- Accounts
 - GitHub: [@hayat01sh1da](#)
 - X: [@hayat01sh1da](#)
 - LinkedIn: [@hayat01sh1da](#)
 - Speaker Deck: [@hayat01sh1da](#)
 - Docswell: [@hayat01sh1da](#)
- Occupation: Software Engineer
- Things I Am Into
 - Linguistics
 - Singing at Karaoke
 - Listening to Music
 - Watching Movies
 - Cat Feeding



Licences / Certifications

- English
 - TOEIC® Listening & Reading 915: Certified on December 2019
- Engineering
 - Information Security Management: Certified on November 2017
 - Applied Information Technology Engineer: Certified on June 2017
 - Fundamental Information Technology Engineer: Certified on November 2016
 - IT Passport: Certified on April 2016
- Others
 - Abacus 2nd Class: Certified on June 2002
 - Mental Arithmetic 3rd Class: Certified on February 2001

Skill Stack

- Languages
 - Japanese: Native Proficiency
 - English: Full professional Proficiency
- Development
 - **Backend: Ruby(Ruby on Rails), Python**
 - Frontend: TypeScript + React.js, TypeScript +Vue.js
 - **Database: MySQL, PostgreSQL, MongoDB**
 - **Architecture: Monolith, Modular Monolith**
 - Hosting: AWS ESK
 - Versioning: Git, GitHub
 - **CI/CD: GitHub Actions, ArgoCD**
 - Monitoring: Datadog, Sentry

Work Histories

Period	Genre	Position	Jobs
Aug 2021 - Present	SaaS(Educational Service)	Software Engineer	• Backend Development • Frontend Development • CI/CD Maintenance • Weekly Release Manager • Troubleshooting • Documentation • Technical Debt Resolution • Tech Blog Entries • LT Speaker
Feb 2020 - Dec 2020	SaaS(Chatbot Platform)	Backend Engineer	• Backend Development • Documentation • Technical Debt Resolution • Inspection of Alternative API

Work Histories

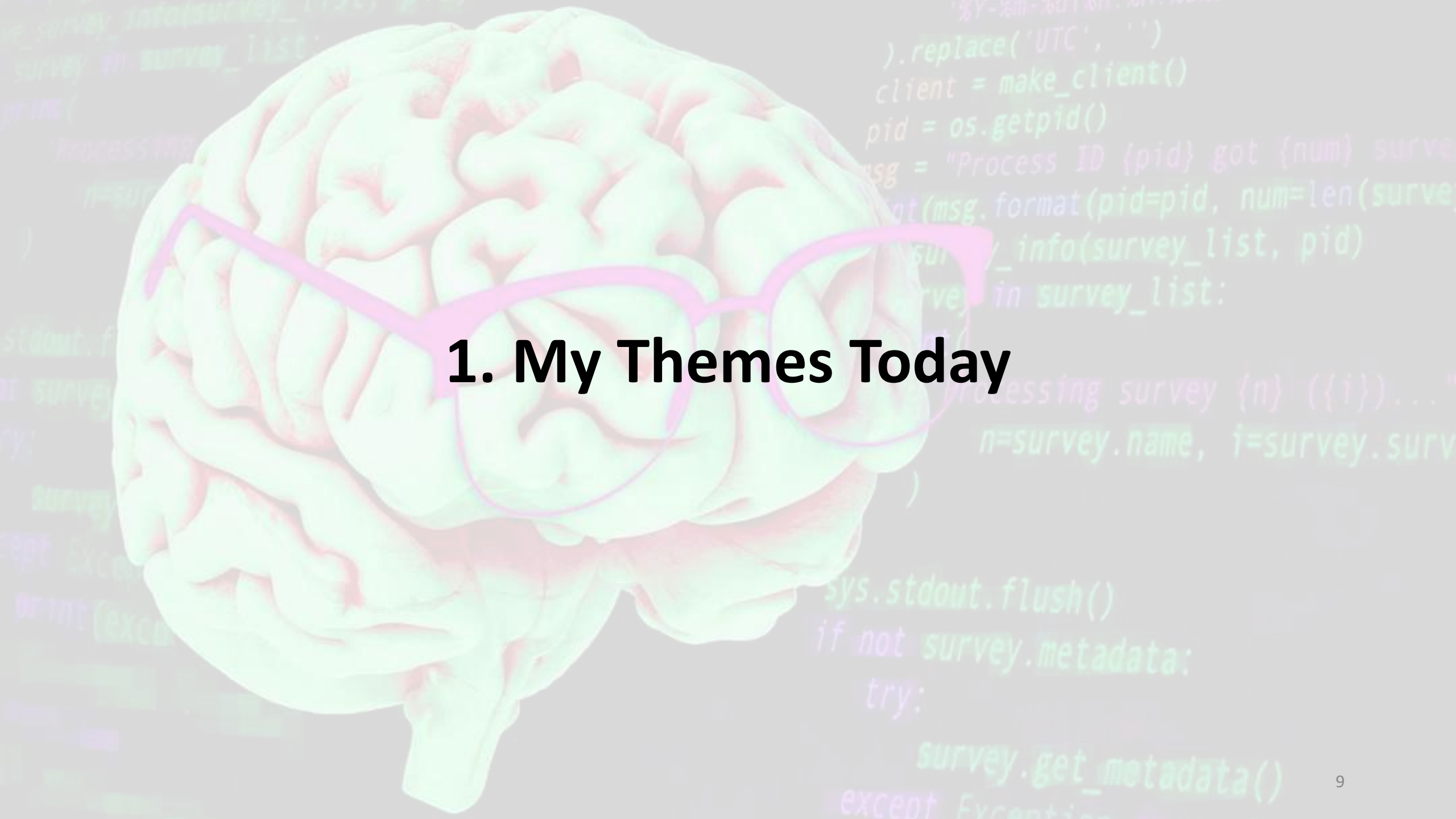
Period	Genre	Position	Jobs
Jun 2018 - Jan 2020	Contract-basis Developer	Software Engineer	• Backend Development • Frontend Development • Troubleshooting • Quality Assurance • Tech Blog Entries
Apr 2016 - Jan 2018	System Engineering Service	System Engineer	• Corporate Account Admin • Windows Server Maintenance • Security Admin Assistant • Internationalisation / Localisation

International-Exchange Activities

- Activities in University
 - English Linguistics Seminar(Focusing on Mass Media English)
 - International-Exchange Clubs(The 2nd Year)
 - [International-Exchange Programmes conducted by Japan Cabinet Office](#)(2013 - 2016)
 - Japanese Linguistics Course(The Final Year)
- Overseas Life Experience
 - Working Holiday in Australia(April 2014 - March 2015)
 - Language School for 1 month in Sydney
 - Work for 6 Months in [Hamilton Island Resort](#)
 - Volunteering for 1 Month as Assistant Teacher of Japanese Language at [St Ives High School](#) in NSW
- Other Activities
 - Keep Everyday Journal in English (April 2014 - Present)
 - [Sunrise Toastmasters Club](#)(February 2017 - March 2018)
 - [Vital Japan](#)(January 2018 - July 2019, October 2022 - February 2023)
 - Self Learning and Training of English Language
 - Participation Ruby-Related Tech Conferences

Agenda

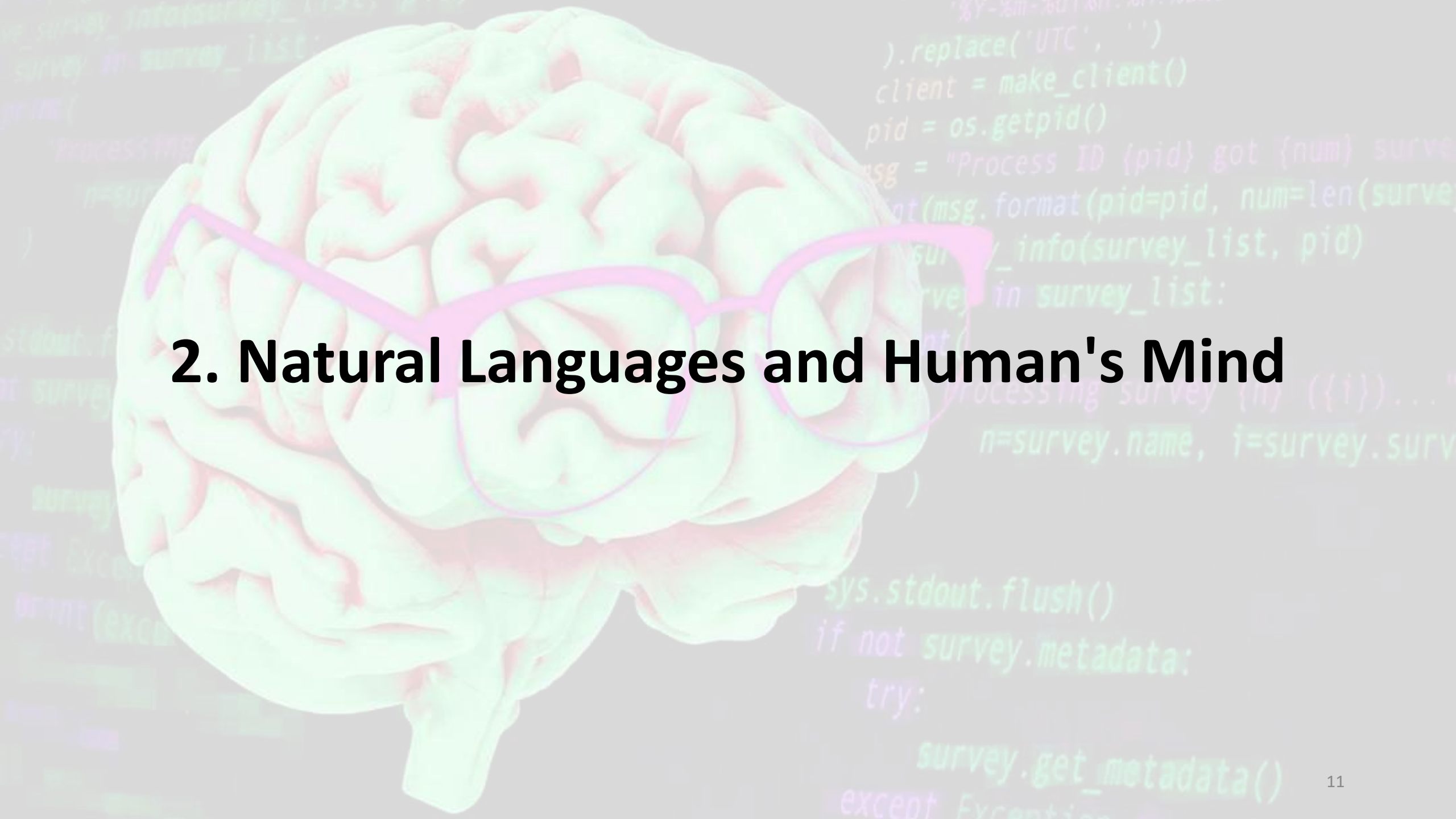
1. [My Themes Today](#)
2. [Natural Languages and Human's Mind](#)
3. [Programming Languages and Human's Mind](#)
4. [Ruby vs. Xxkspeak](#)
5. [Conclusion](#)
6. [References](#)



1. My Themes Today

1. My Themes Today

1. To **NOT** delicate myself to prosperity of Ruby at all.
2. To earnestly discuss something **NEITHER** practical **NOR** technical.
3. To take audience to a short journey of a **linguistic thought experiment**.



2. Natural Languages and Human's Mind

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Natural Language and Communication

We are involved in these activities via natural language as the interface.

Category of Communication	Activity	Act / Factor
External Communication = Social Activity	Exchange Information <u>with People</u>	• Asking Questions • Making Assertions
	Regulate and Coordinate Behaviours <u>of People</u>	• Making Requests • Giving Orders • Making Offers • Making Promises
Internal Communication = Mental Activity	Organise Knowledge <u>in Oneself</u> = Classification of the Objects of the World	• Living vs. Non-Living • Plants vs. Animals • Humans vs. Other Animals
	Reason and Draw an Inference <u>in Oneself</u> = Read between the Lines	• What it means • What it implies

2. Natural Languages and Human's Mind

Newspeak in *Nineteen Eighty-Four* written by George Orwell

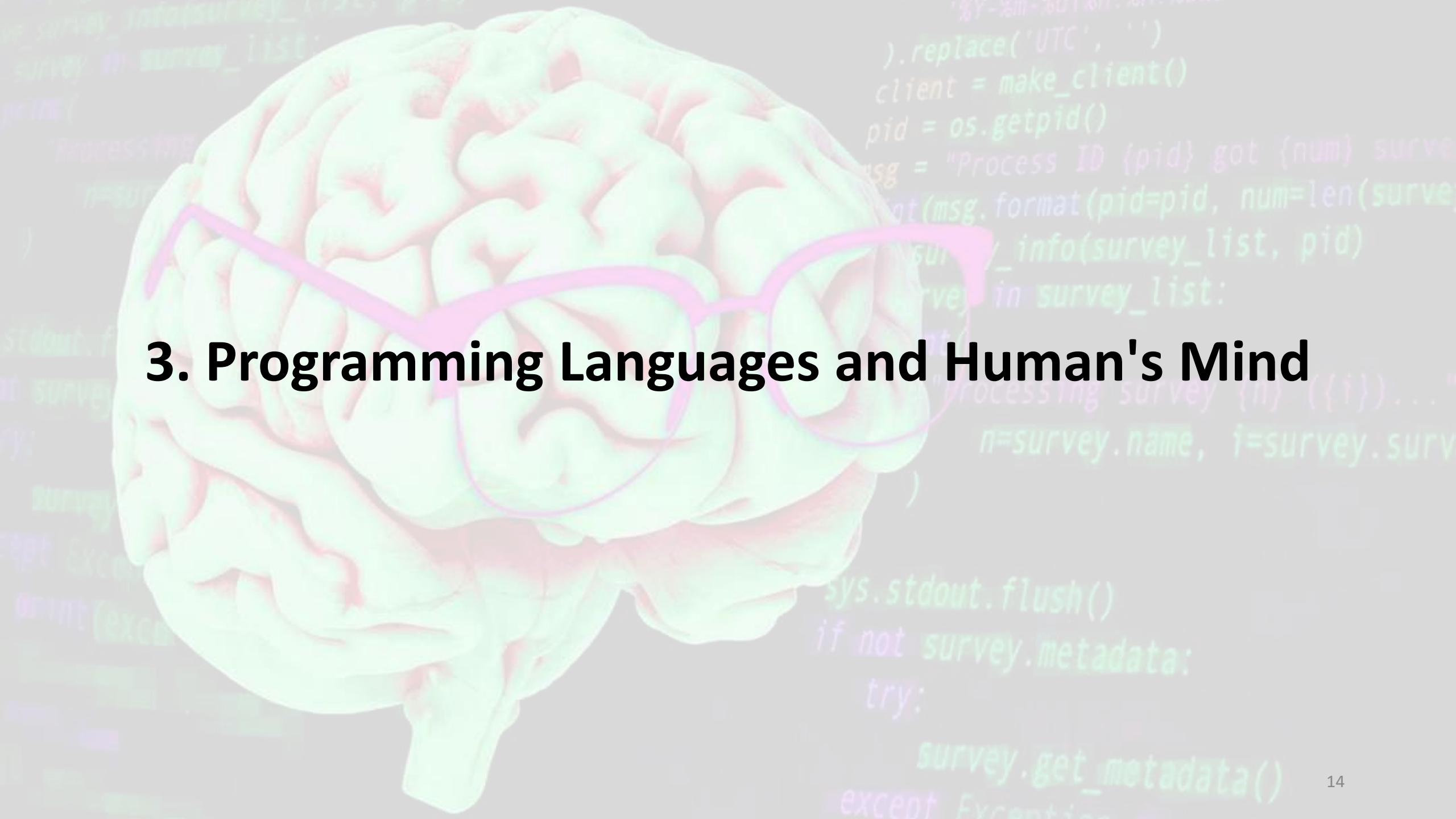
The language was designed and partly enforced in Oceanic society in the novel.

The main purpose was to **regulate exterminate anti-social political thoughts by eliminating words** related to them along with the concepts.

That is to say, we also would lose the corresponding concept if we lost a word.

Let's discuss differences between **Oldspeak**(Present-Day English) and **Newspeak**.

Attribute	Oldspeak(Present-Day English)	Newspeak
Vocabulary and Expressions	Richer every second	Poorer every year
Regulation of Human's Mind	Looser	Stricter
Flexibility of Expressions	Extraordinarily High	Not Flexible at all
Ambiguity	Less	More
Cognitive Load	Lower(Low Context)	Higher(High Context)
Unanimity of Expressions	Hard	Easy



3. Programming Languages and Human's Mind

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Programming Language and Communication

We are involved in these activities via programming language as the interface.

Category of Communication	Activity	Act / Factor
External Communication = Social Activity	Exchange Information <u>with People</u>	• Asking Questions • Making Assertions
	Regulate and Coordinate Behaviours <u>of not just Computers but People</u>	• Making Requests • Giving Orders • Making Offers • Making Promises
Internal Communication = Mental Activity	Organise and Apply Knowledge <u>in Oneself</u> = Design and Implementation	• Controller vs. Model • ActiveRecord vs. CQRS • Monolith vs. Microservice
	Reason and Draw an Inference <u>in Oneself</u> = Read between the Codes	• What it implements • What requirements it implies

3. Programming Languages and Human's Mind

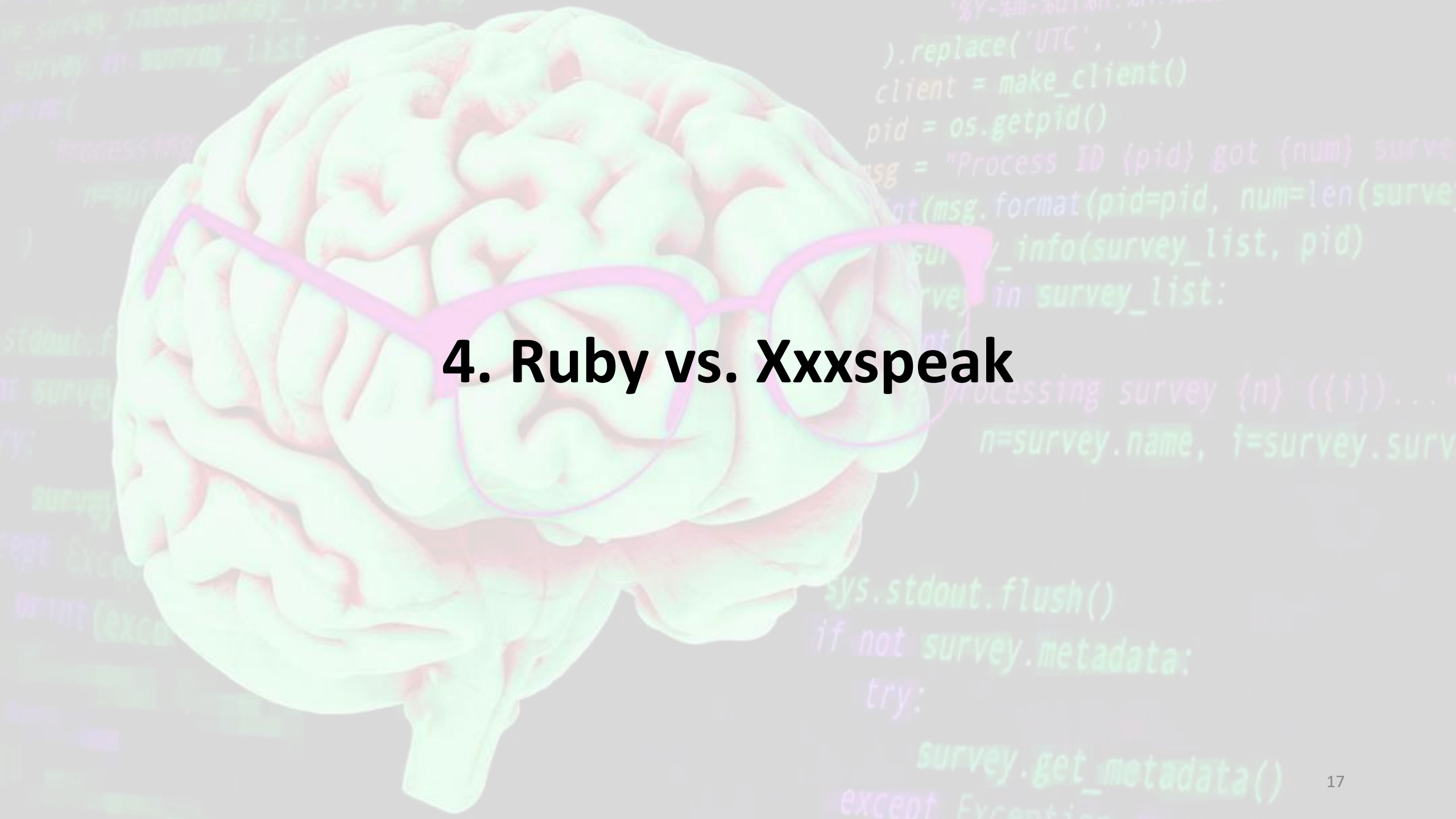
Ruby created by Mr. Yukihiro Matsumoto(generally called Matz)

Ruby is outstanding in its [Linguistic Cognitive Consideration](#) learned from Perl.

The feature provides Rubyists with a wonderful coding experience.

Let's discuss differences between Ruby and Python, another OOP.

Attribute	Ruby	Python
Vocabulary and Expressions	Rich	Limited
Regulation of Human's Mind	Looser	Stricter
Flexibility of Expressions	Higher	Lower
Ambiguity	Less(Naming convention)	Less(Language design)
Cognitive Load	Hard to Evaluate(High and/or Low)	Hard to Evaluate(High and/or Low)
Unanimity of Expressions	Hard(Easier with tool introduction)	Easy



4. Ruby vs. Xxxspeak

4. Ruby vs. Xxkspeak

Ruby vs. Newspeak

The comparison below shows the less mutual features.

Attribute	Ruby	Newspeak
Vocabulary and Expressions	Rich	Poorer every year
Regulation of Human's Mind	Looser	Stricter
Flexibility of Expressions	Higher	Not Flexible at all
Ambiguity	Less(Naming convention)	More
Cognitive Load	Hard to Evaluate(High and/or Low)	Higher(High Context)
Unanimity of Expressions	Hard(Easier with tool introduction)	Easy

4. Ruby vs. Xxkspeak

Ruby vs. Oldspeak

The comparison below shows the more mutual features because both of them think highly of **Convention over Configuration**.

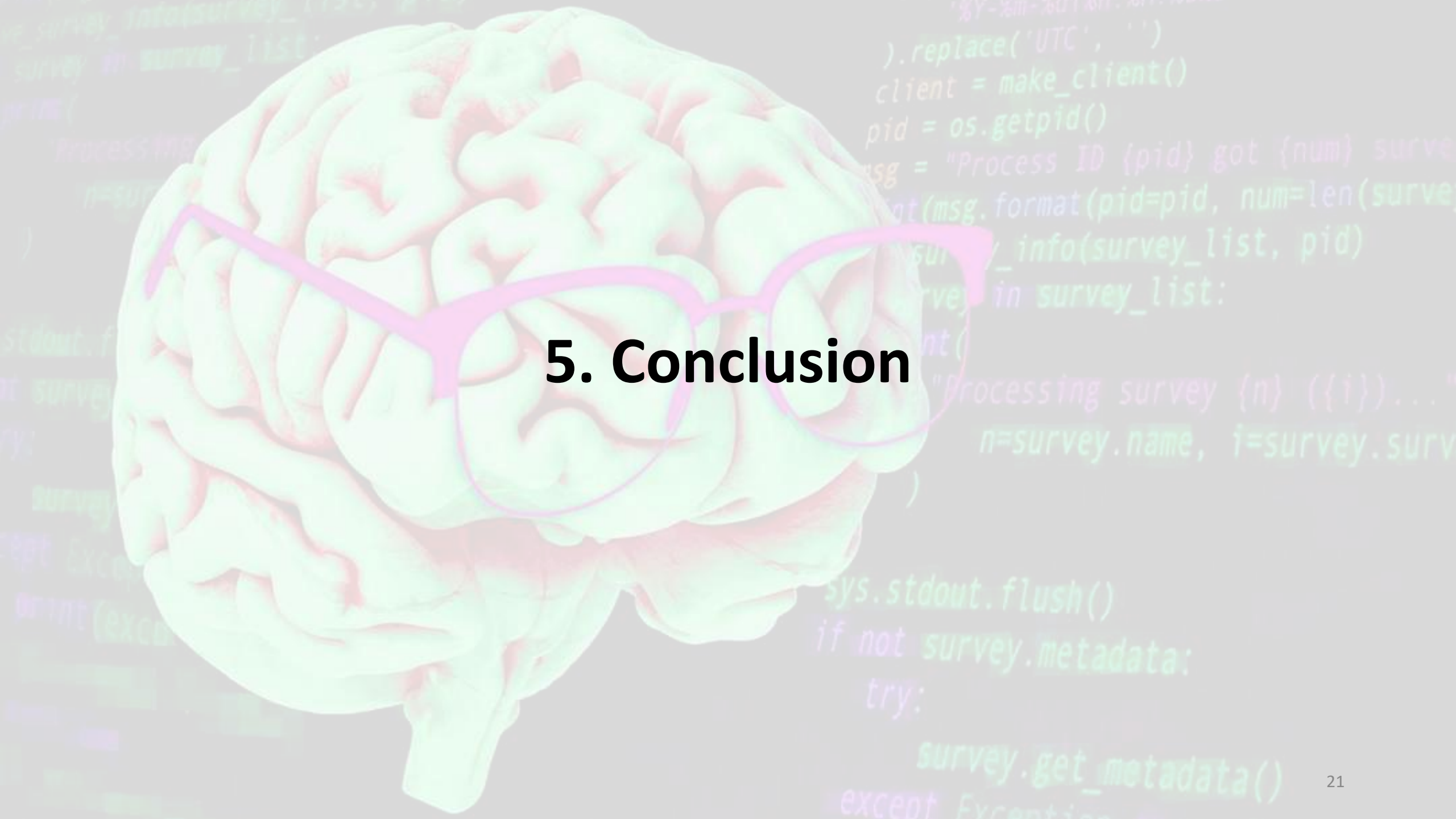
Attribute	Ruby	Oldspeak(Present-Day English)
Vocabulary and Expressions	Rich	Richer every second
Regulation of Human's Mind	Looser	Looser
Flexibility of Expressions	Higher	Extraordinarily High
Ambiguity	Less(Naming convention)	Less
Cognitive Load	Hard to Evaluate(High and/or Low)	Lower(Low Context)
Unanimity of Expressions	Hard(Easier with tool introduction)	Hard

4. Ruby vs. Xxxspeak

[Appendix] Python vs. Newspeak

The comparison below shows the more mutual features because both of them are think highly of **Configuration over Convention**.

Attribute	Python	Newspeak
Vocabulary and Expressions	Limited	Poorer every year
Regulation of Human's Mind	Stricter	Stricter
Flexibility of Expressions	Lower	Not Flexible at all
Ambiguity	Less(Language design)	More
Cognitive Load	Hard to Evaluate(High and/or Low)	Higher(High Context)
Unanimity of Expressions	Easy	Easy



5. Conclusion

5. Conclusion

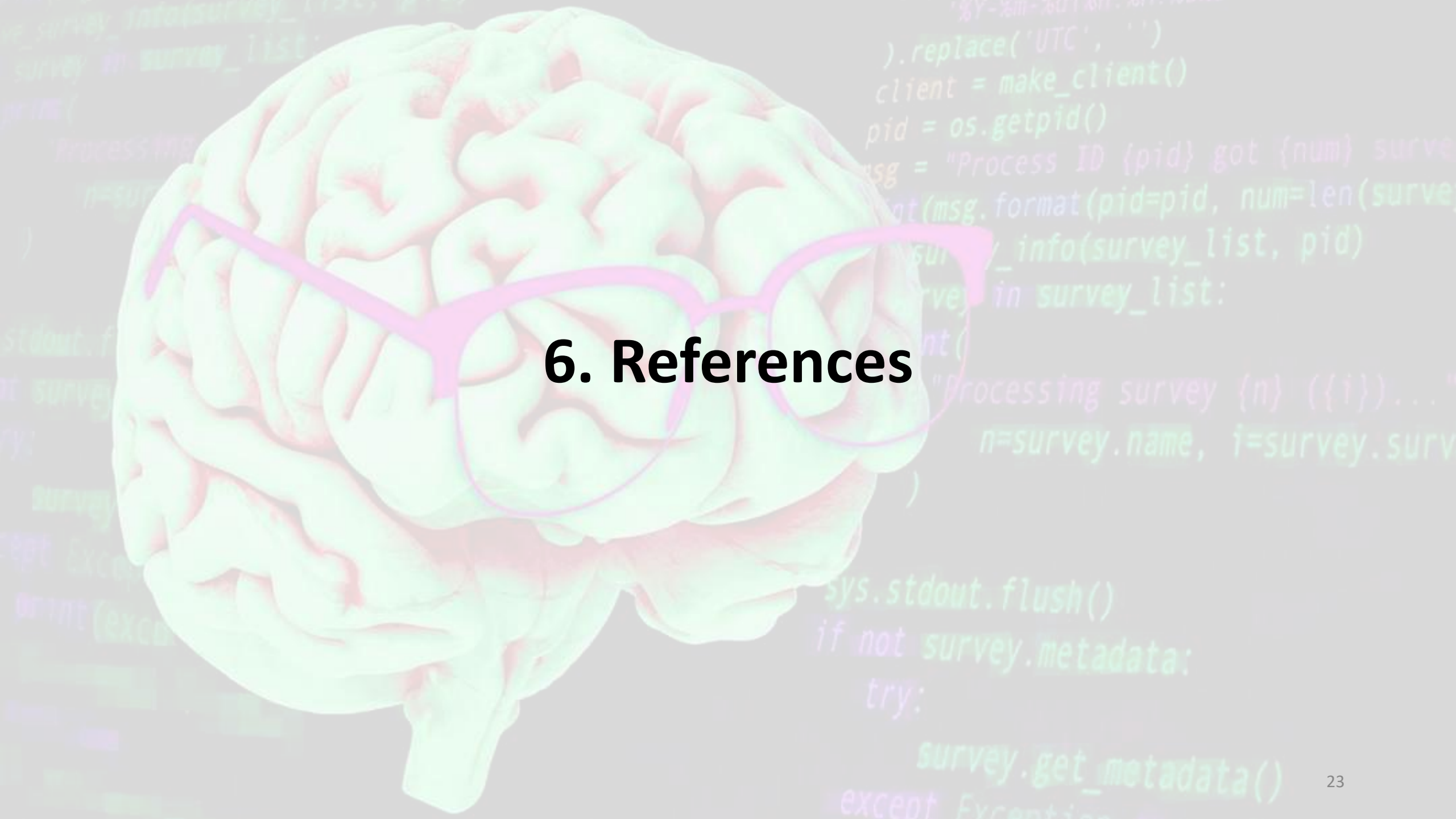
The following principles apply to both of natural and programming languages.

1. **Rich** vocabulary in terms of communication interface provides

- **higher** flexibility of expressions thanks to **looser** regulation of human's mind
- **higher** cognitive load due to too **rich** idioms to properly handle
- **more** costs for arrival at the agreement to define ubiquitous expressions within the same community

2. **Poor** vocabulary in terms of communication interface provides

- **lower** flexibility of expressions owing to **stricter** regulation of human's mind
- **lower** cognitive load thanks to so **poor** idioms as to easily handle
- **less** costs for arrival at the agreement to define ubiquitous expressions within the same community



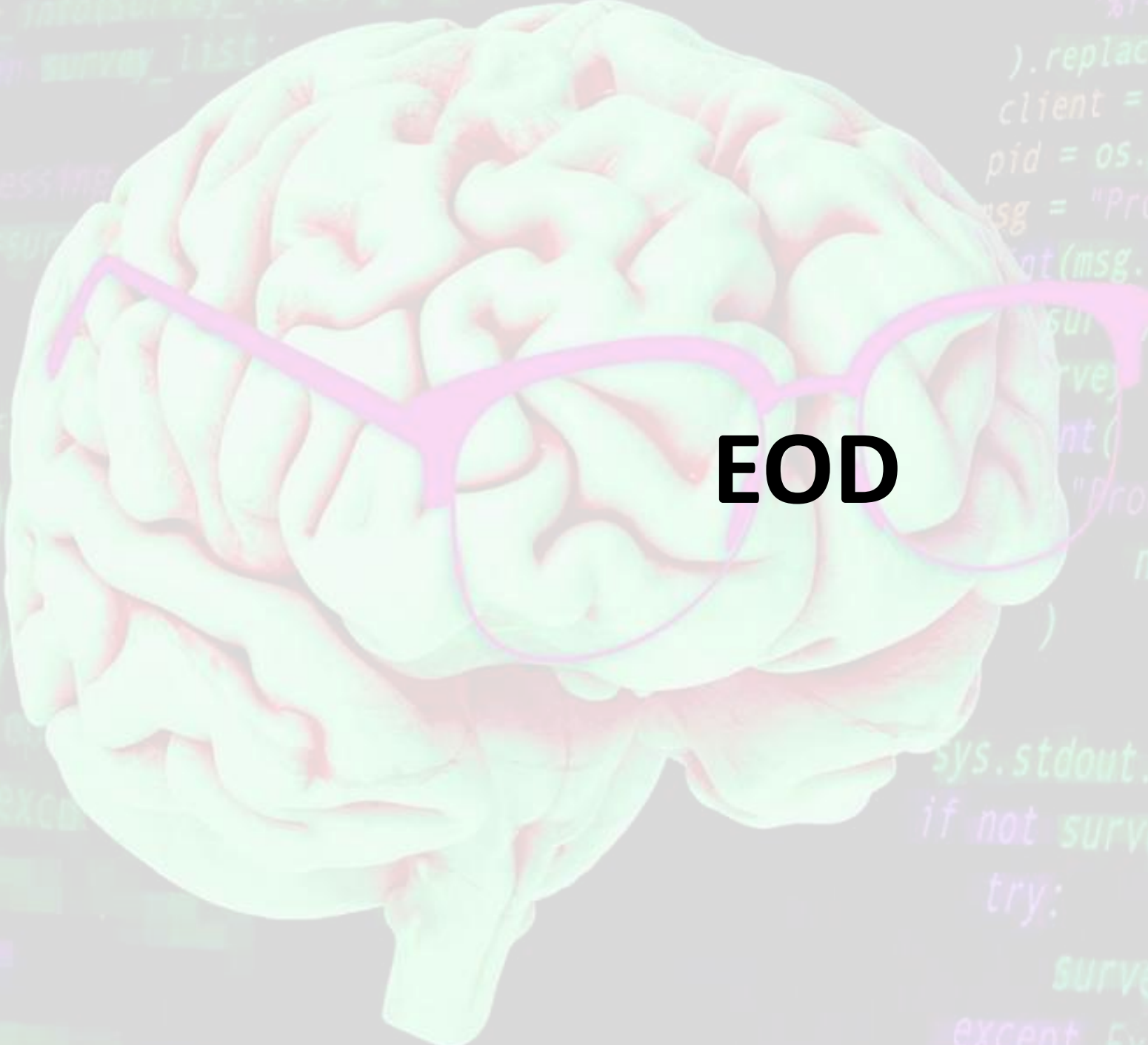
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6. References

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- Hayato Ishida, [*Economy of Efforts*](#), Tokyo, 2025
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- Hayato Ishida, [*Road to Ruby for A Linguistics Nerd*](#), Tokyo, 2025
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